

EDUGENIE – GOOGLE GEMINI POWERED LEARNING ASSISTANT

1. Introduction

EduGenie is an AI-powered learning assistant designed to help students understand and study educational topics more effectively. The system uses **Google Gemini** for AI-based question answering, explanations, quizzes, summarization, and personalized learning recommendations.

2. Problem Statement

Students often need quick explanations, practice questions, summaries, and learning guidance while studying. Traditional learning resources may require searching through multiple sources. EduGenie provides these learning-support features through a single web application.

3. Project Objectives

- Provide AI-based answers to student questions.
- Explain concepts in a simple and understandable way.
- Generate short multiple-choice quizzes.
- Summarize educational passages.
- Provide personalized learning recommendations.
- Develop a simple and user-friendly web interface.

4. Proposed Solution

EduGenie combines a **FastAPI backend**, **Jinja2 frontend**, and **Google Gemini AI services**. Users enter their learning requirements through the web interface, and the backend processes the request using the appropriate AI functionality before returning the result.

5. System Architecture

The basic architecture is:

User → Web Interface → FastAPI Backend → Gemini AI → Response

The application provides separate API endpoints for Q&A, concept explanation, quiz generation, summarization, and learning recommendations.

6. Technologies Used

Technology	Purpose
Python	Application development
FastAPI	Backend and API development
Jinja2	Web page rendering
HTML/CSS	Frontend interface
Google Gemini AI/LLM functionality	
Pytest	Automated testing
Git/GitHub	Version control

7. Main Features

7.1 Question Answering

Users can ask educational questions and receive AI-generated answers.

7.2 Concept Explanation

The system explains topics according to the learner's requested level.

7.3 Quiz Generation

EduGenie generates a short three-question multiple-choice quiz from provided content.

7.4 Text Summarization

Long educational passages can be converted into concise summaries.

7.5 Learning Recommendations

The system provides personalized learning recommendations based on the selected topic and learner level.

8. System Modules

1. **Q&A Module** – Handles educational questions.
2. **Explanation Module** – Provides concept explanations.
3. **Quiz Module** – Generates MCQ-based quizzes.
4. **Summarization Module** – Creates concise summaries.
5. **Learning Recommendation Module** – Suggests learning paths.
6. **Web Interface Module** – Provides user interaction.

9. API Endpoints

Endpoint	Function
/qa	Question answering
/explain	Concept explanation
/quiz	Quiz generation
/summarize	Text summarization
/learn/recommendations	Learning recommendations
/health	Application health check
/docs	API documentation

These endpoints are implemented using the FastAPI application.

10. AI Integration

The project uses the current **Google GenAI Python SDK** with a configurable Gemini model. The default model configured by the project is **Gemini 2.5 Flash**.

An optional local explanation model can also be enabled. When enabled, the application can use a local LaMini-based model, with Gemini available as a fallback.

11. Testing

The application was tested using:

- Functional testing
- API testing
- Input validation
- Integration testing
- Frontend testing
- AI service testing
- Automated testing using Pytest

The project tests mock Gemini requests, allowing automated tests to run without making paid API calls.

12. Security Considerations

The Gemini API key is stored using environment variables and should not be committed to GitHub. The project also recommends additional security measures such as authentication, rate limiting, HTTPS, request-size limits, and proper secret management for production deployment.

13. Results

EduGenie successfully provides AI-powered educational assistance through a web-based interface. The implemented modules support question answering, explanations, quizzes, summaries, and learning recommendations.

14. Advantages

- Simple and easy to use.
- Provides multiple learning features in one application.
- Uses generative AI for personalized responses.
- Supports different learning requirements.
- Provides API documentation through FastAPI.
- Easy to extend with additional educational features.

15. Future Enhancements

Future versions can include:

- Voice-based learning
- Multilingual support
- Mobile application
- Student progress tracking
- Gamification
- Adaptive learning
- Image/PDF-based learning
- LMS integration
- Collaborative learning features

16. Conclusion

EduGenie demonstrates how generative AI can be integrated into an educational web application to support students with learning and revision tasks. The project combines

FastAPI, a web interface, and Google Gemini to provide multiple AI-powered learning features in a single system.

Project Name: EduGenie – Google Gemini Powered Learning Assistant

Repository: [deepukutty9812-ai/EduGenie-AI](https://github.com/deepukutty9812-ai/EduGenie-AI)